

Milestone 1: Data visualization

Problematic

Overview

Blue Zones are five geographic regions where people tend to live significantly longer and healthier lives than the global average: Okinawa (Japan), Sardinia (Italy), Nicoya Peninsula (Costa Rica), Ikaria (Greece), and Loma Linda (USA). Researchers studying these areas identified nine lifestyle principles (often referred to as the Blue Zone “Power 9”) that appear to contribute to this exceptional longevity. These principles include four spiritually oriented pillars and five related to daily habits, such as natural movement (walking, gardening etc.), stress reduction, the 80% Rule (stopping eating before feeling full), a predominantly plant-based diet, and moderate wine consumption.

Inspired by these findings, our project explores a broader question: **Which countries exhibit measurable lifestyle patterns similar to Blue Zone principles, and how strongly do these patterns correlate with life expectancy?**

To investigate these relationships, we will develop an interactive and playful data-driven website that allows users to explore longevity patterns across the world.

Motivation

In a society obsessed with finding the fountain of youth and unnatural ways to extend their lives, Blue Zones seem to have found a secret to naturally attain such longevity. Our motivation for this project comes from the fact that Blue Zones remain relatively unknown to the general public. Although this is a fascinating area of research, there has been limited exploration of these longevity patterns at a broader, country-level scale. By analyzing global data and presenting it through an interactive platform, the project aims to make these patterns more accessible and easier to understand.

Target audience

The project is designed for a broad audience and does not require any technical knowledge to explore. Anyone curious about Blue Zones and longevity can engage with the experience. Whether users want to learn about lifestyle factors that may contribute to longer life expectancy, discover which countries align most closely with these lifestyles, or simply explore engaging data visualizations, the platform offers an accessible and interactive way to learn.

Dataset

To investigate our key research question, it was essential to identify datasets that appropriately represent each of the selected pillars.

In terms of quality, three of the datasets ([Dietary](#), [Life Expectancy](#), [Wine Consumption](#)) were obtained from *Our World in Data*, a website that provides datasets from scientific studies or organizations such as the World Health Organization (WHO). The dataset on physical

activity and movement was sourced from a study conducted by Stanford University about [Activity Inequality](#). Finally, due to the availability of data, a choice was made to replace stress evaluation in countries by happiness level. A dataset from [World Happiness Report](#) has been used to analyse such emotion. The credibility of these sources supports the reliability and validity of our visualizations.

While these sources are reliable and widely used, they present limitations. For instance, happiness is not a direct measure of stress, physical activity data covers only 111 countries, and wine consumption does not fully capture alcohol habits. These constraints must be considered when interpreting results.

Preprocessing includes standardizing country names, filtering some data to 1950–2023, computing derived metrics (e.g., plant-based calorie ratios), and merging datasets into a unified structure.

Exploratory Data Analysis

Our exploratory analysis highlights several key insights:

Life Expectancy Dataset

For the Life Expectancy dataset, we filtered the timeframe to 1950-2023 for consistency. We found that countries containing Blue Zone regions average a higher life expectancy of 82.07 years compared to 73.9 years in other countries.

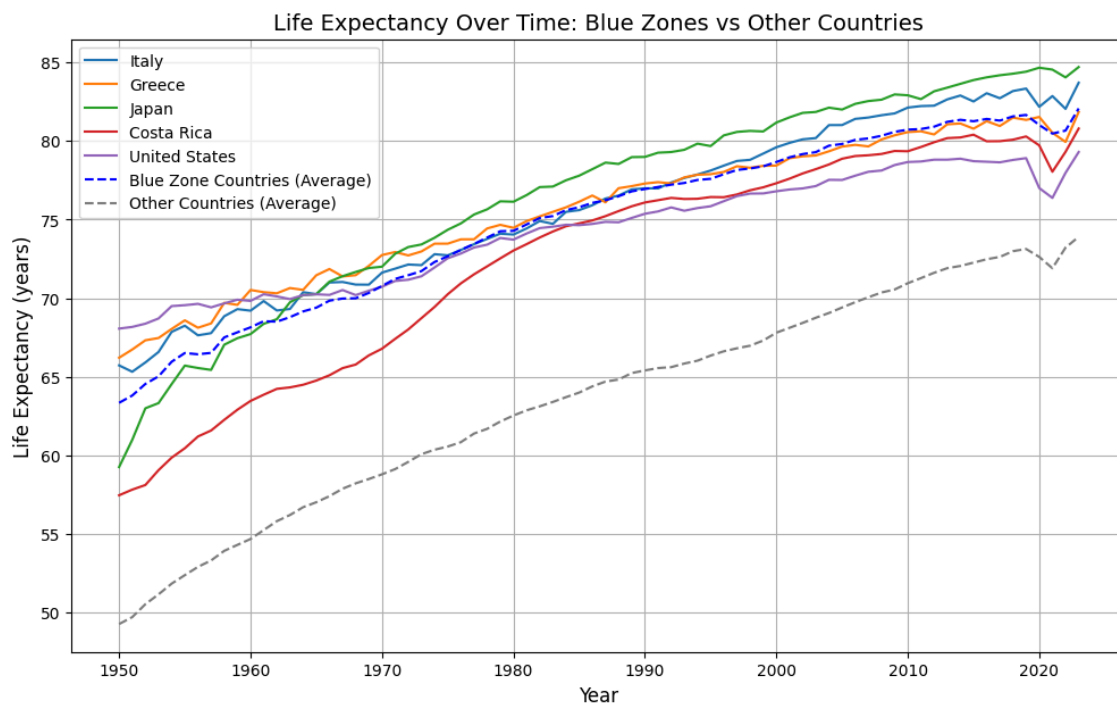


Figure 1: Life expectancy evolution (1950-2023)

A heatmap of the top 30 countries further illustrates the evolution of life expectancy over time (ranked based on their life expectancy in 2023).

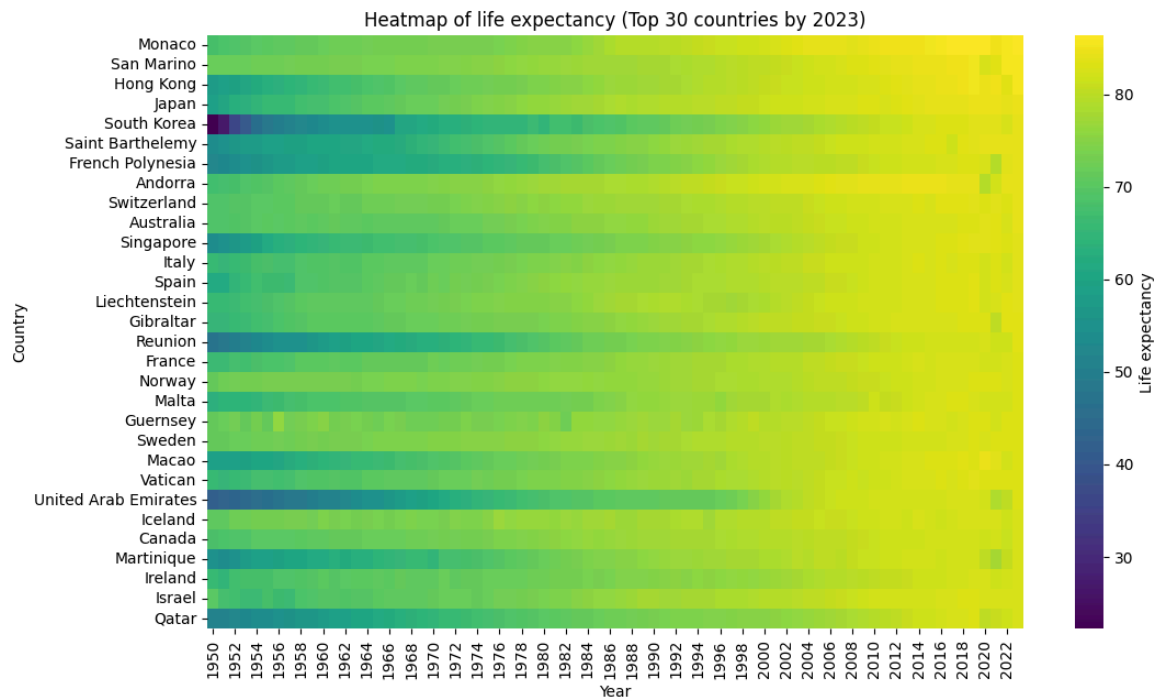


Figure 2: *Heatmap of life expectancy for the top 30 countries by 2023*

Number of Steps Dataset

Exploring the "moving naturally" pillar via daily steps (covering 111 countries), the distribution is slightly right-skewed. We observed a factor of two differences between the lowest (El Salvador) and highest (Hong Kong) daily steps, but Blue Zone countries do not particularly stand out, indicating that this factor alone is insufficient to justify the higher longevity in these zones.

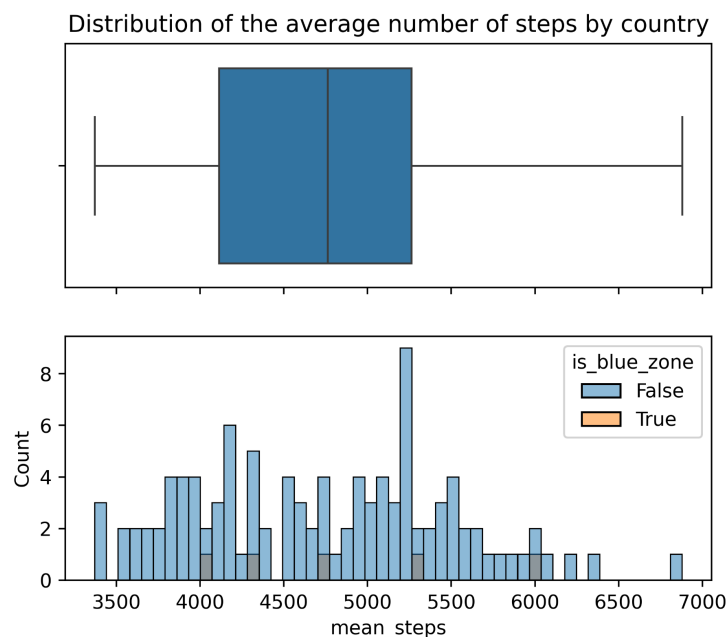


Figure 3: *Distribution of the average number of daily steps across 111 countries*

	Entity	mean_steps
31	El Salvador	3370.078247
40	Honduras	3382.804431
76	Pakistan	3413.878452
45	Indonesia	3513.438387
100	Trinidad	3520.292596

	Entity	mean_steps
51	Japan	6009.770680
103	Ukraine	6106.926110
21	China	6188.650349
60	Macau SAR China	6346.895307
41	Hong Kong SAR China	6879.679031

Figure 4 & 5: *Top 5 countries with respectively lowest and highest average number of daily steps*

Wine Consumption Dataset

As for wine consumption, the research showed that moderate drinkers tend to outlive non-drinkers. The apparent sweet spot is around 1-2 glasses per day. The countries that follow this pattern are shown below. However, surprisingly, from Blue Zone countries only Italy appears to have followed this pattern in 2020.

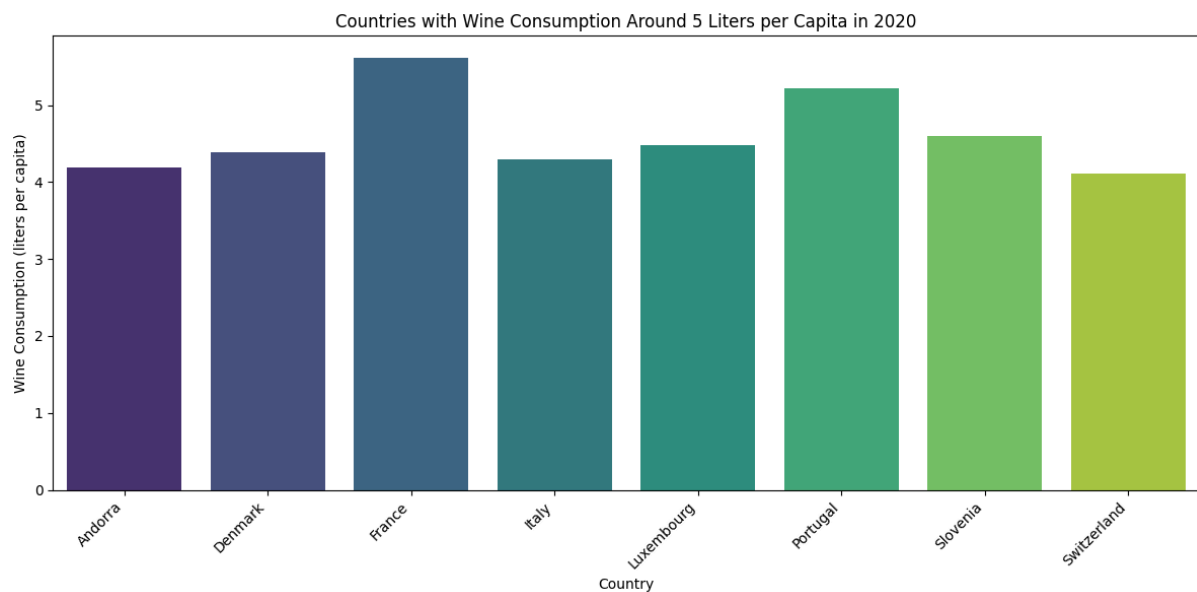


Figure 6: *Countries with wine consumption around 5 liters per capita in 2020*

Dietary Composition by Country and Happiness Datasets

Finally, analyzing Dietary Composition revealed a major paradox. While the US has been leading in total caloric intake for years, the countries closest to the theoretical plant-based Blue Zone diet are in Africa (e.g. Nigeria at 0.93 and Madagascar at 0.92 plant-based ratios).

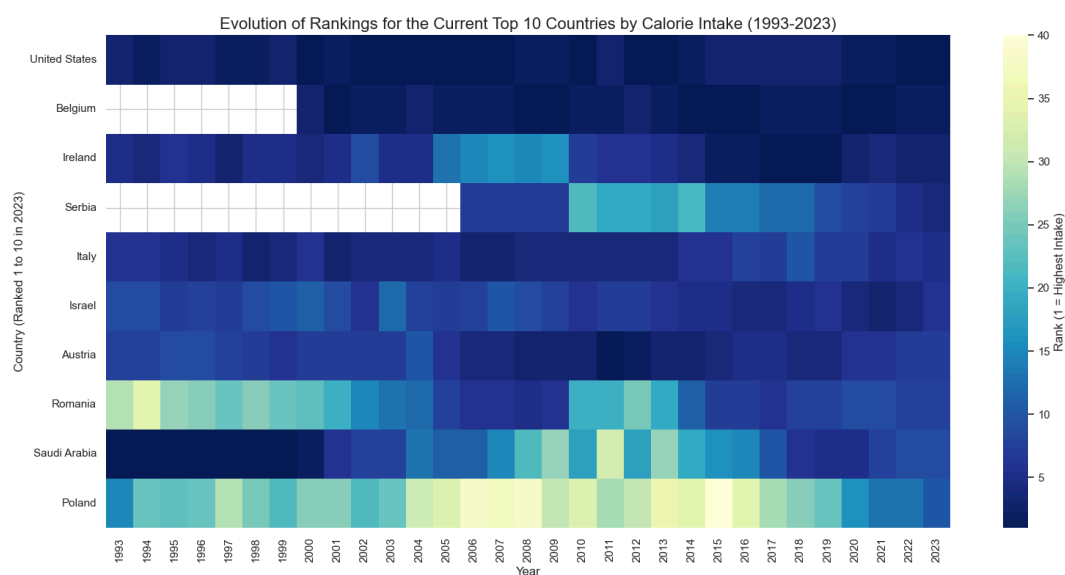


Figure 7: *Evolution of the ranking of the current top 10 countries by calorie intake between 1993 and 2023*

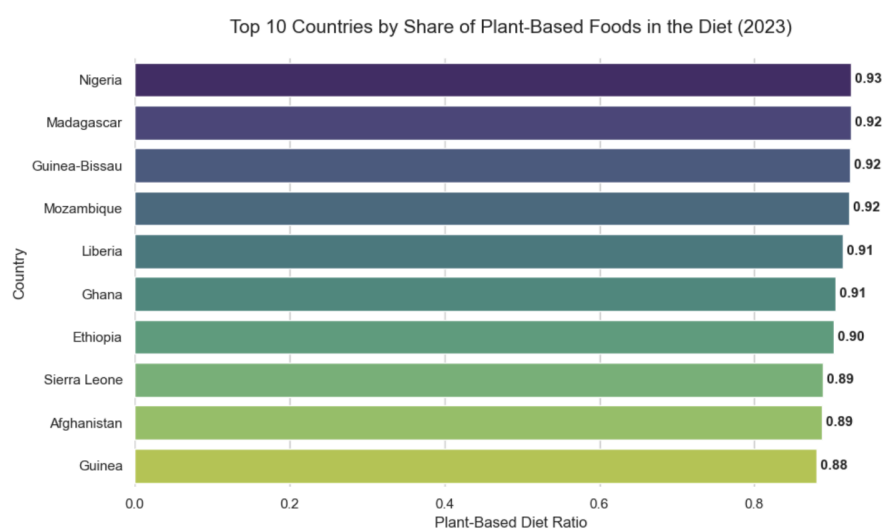


Figure 8: *Top 10 countries by share of plant-based foods in the diet (2023)*

Yet, these nations face some of the world's lowest life expectancies and happiness scores (e.g., Nigeria ranking 207th and Afghanistan 143rd respectively).

Entity	Life expectancy	Rank
Afghanistan	66.0346	177
Ethiopia	67.3150	167
Ghana	65.4979	181
Guinea	60.7405	200
Guinea-Bissau	64.0849	185
Liberia	62.1625	194
Madagascar	63.6317	190
Mozambique	63.6111	191
Nigeria	54.4623	207
Sierra Leone	61.7858	196

Country name	Rank
Afghanistan	143
Ethiopia	130
Ghana	120
Guinea	97
Liberia	121
Madagascar	123
Mozambique	90
Nigeria	102
Sierra Leone	140

Figure 9 & 10: *Life expectancy and happiness rankings for the top 10 countries with the highest plant-based food ratio in 2023*

These results show that single pillars fail to explain longevity. They instead motivate a multidimensional approach combining diet, happiness, physical activity and wine consumption. The goal is to develop a multidimensional "Blue Zone Index" : $BZI = w_1 \cdot \text{Diet} + w_2 \cdot \text{Steps} + w_3 \cdot \text{Wine} + w_4 \cdot \text{Happiness}$, with weights chosen based on empirical analysis. This index aims to identify "Turquoise Countries" that share some aspects of life with Blue Zones.

Related Work

What others have already done with the data?

Our datasets have been used to conduct several analyses, including:

- [World Happiness Report](#): Analyzes the World Happiness Report to explore factors influencing countries' happiness and predict happiness scores.
- [Activity Inequality](#): Studies how physical activity levels vary across populations and how socioeconomic and geographic factors contribute to activity inequality.
- [Alcohol Consumption](#): Presents analysis on global patterns and trends in alcohol consumption, exploring how consumption varies by country, over time, and in relation to health outcomes.
- [Life Expectancy: What does this actually mean?](#): Explains how life expectancy is calculated, what it measures, and how it should be interpreted when comparing populations and tracking changes over time.

However, since the concept of Blue Zones is relatively new and quantitative data for these regions is limited, they have been studied very little. The only relevant work we were able to find is the [Blue Zones Longevity Analysis](#).

Why is your approach original?

The [Blue Zones Longevity Analysis](#) primarily limits itself to comparing life expectancy trends. Our approach is original because, unlike most datasets which are based on measurable data, there is no publicly available data specifically for Blue Zones. Consequently, our project dynamically focuses on the pillars supposed to be associated with longevity in Blue Zones. Instead of just observing the zones, we build an interactive index to identify which countries around the world exhibit these patterns.

What source of inspiration do you take?

Our work was inspired by the Netflix documentary *Live to 100: Secrets of the Blue Zones*, which sparked our initial interest in exploring this topic in greater depth.

Regarding the design and presentation of our visualizations for the upcoming website, while our choices are not yet finalized, we plan to draw inspiration from old projects, particularly the [OECD Well-being Interactive Map](#).